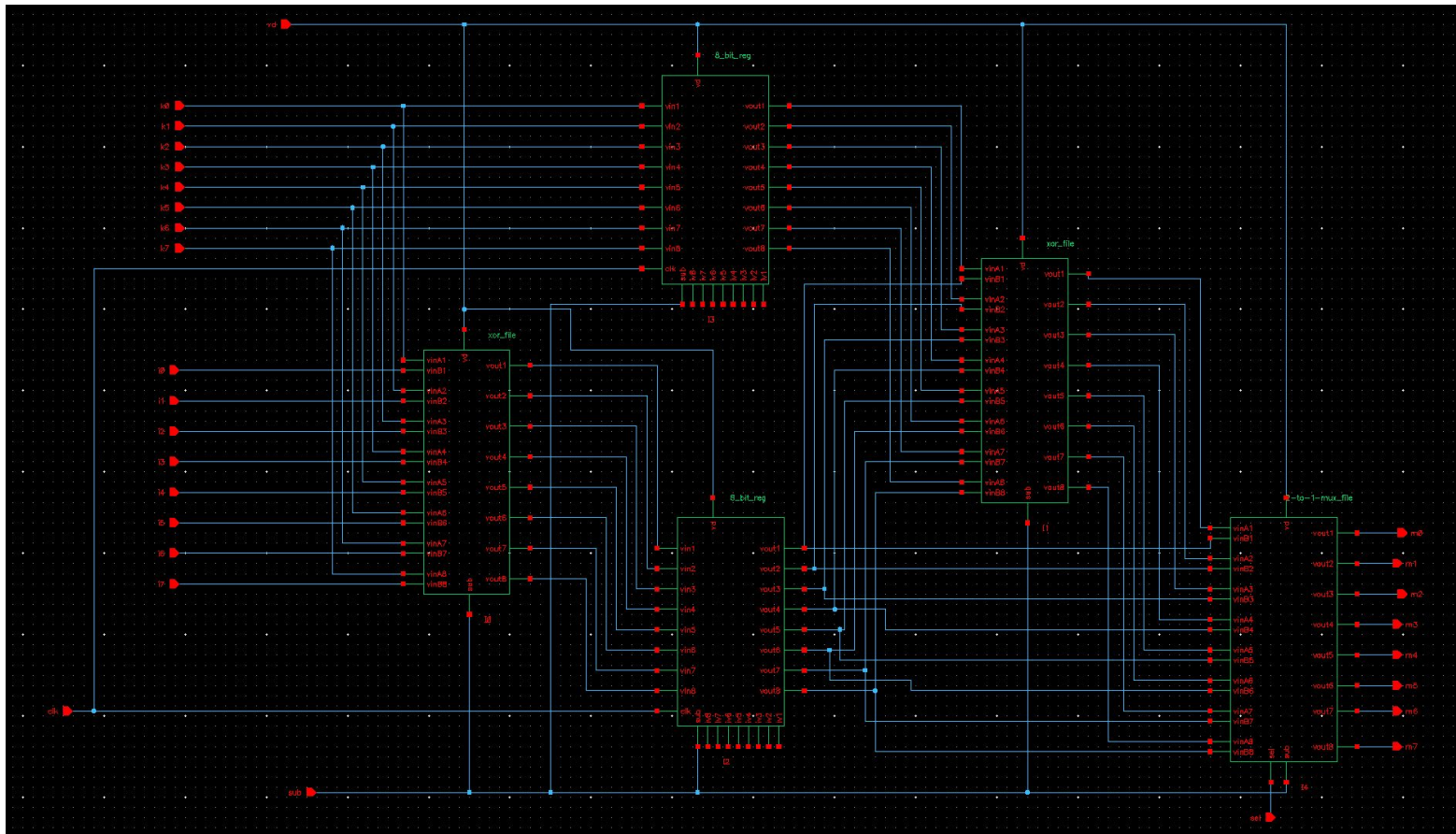




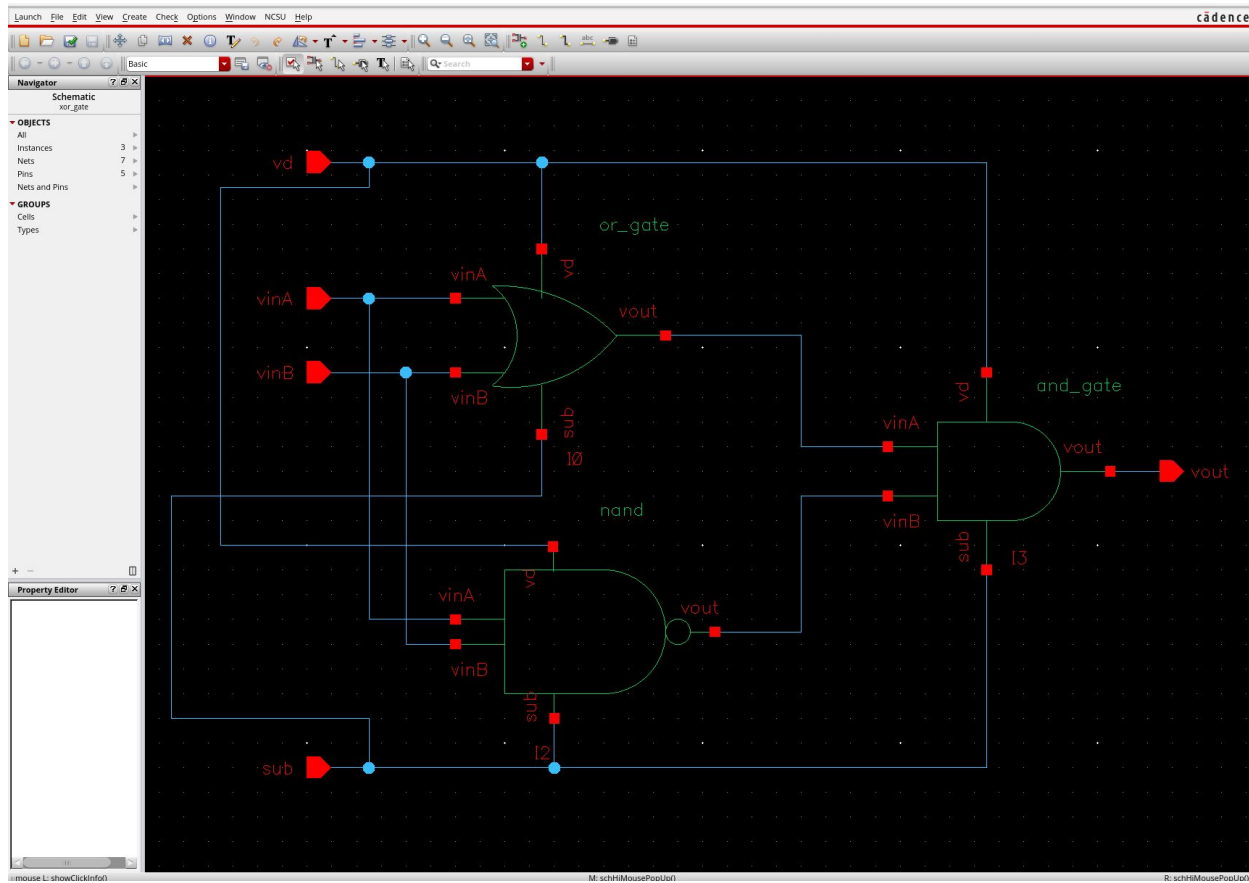
XOR Encryptor/ Decryptor

Project 2 -Schematic and Sim





XOR Encryptor/ Decryptor Circuit - 3 sections

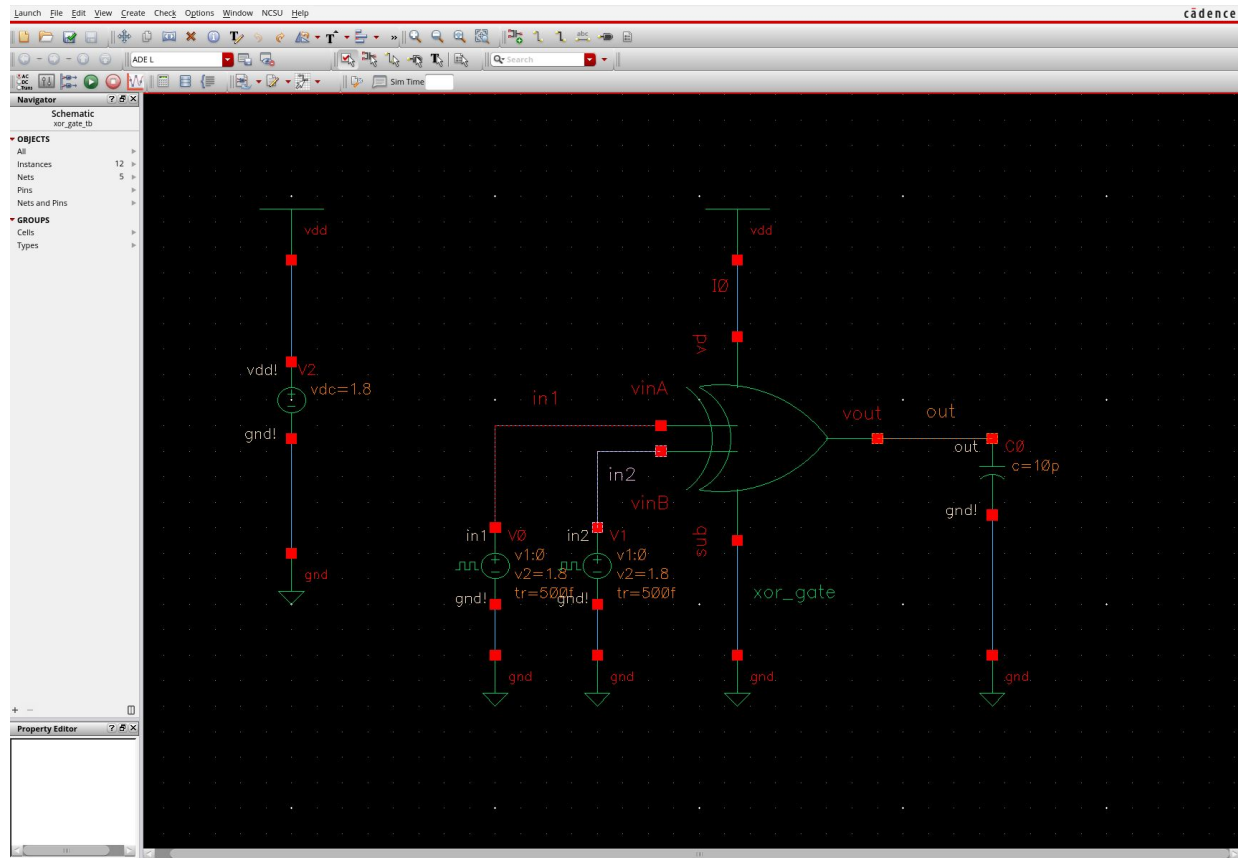


0 AND 0 = 0
0 AND 1 = 0
1 AND 0 = 0
1 AND 1 = 1

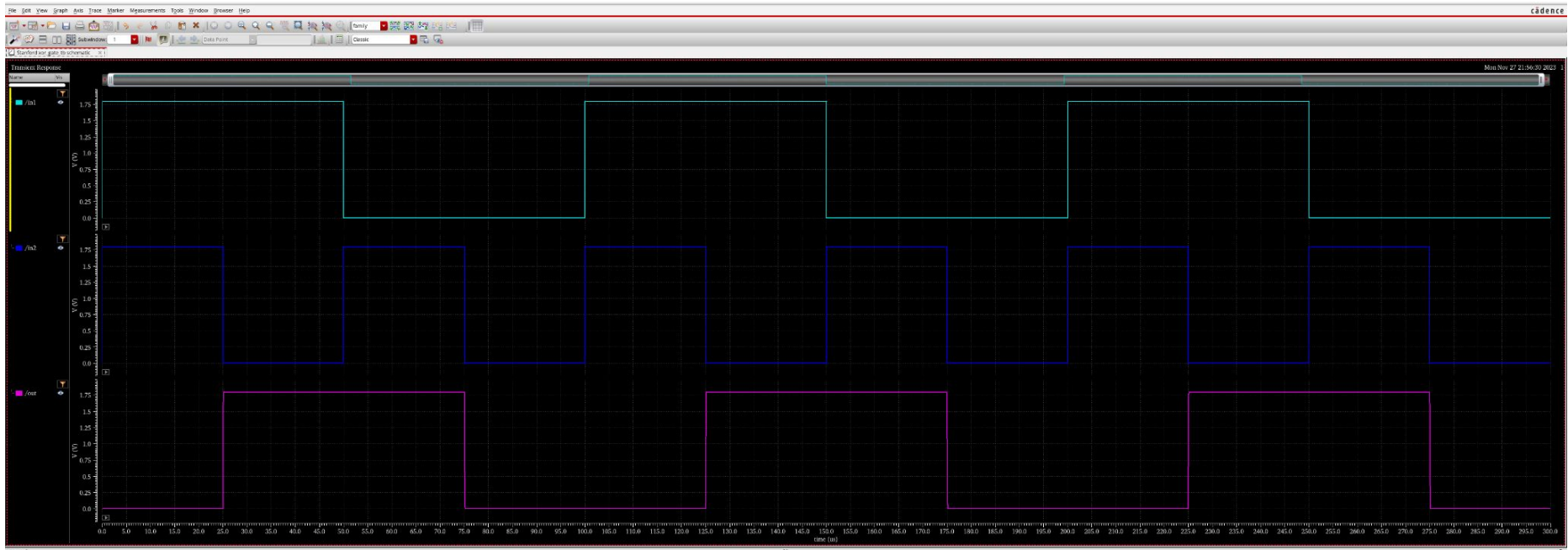
XOR GATE



Encryption

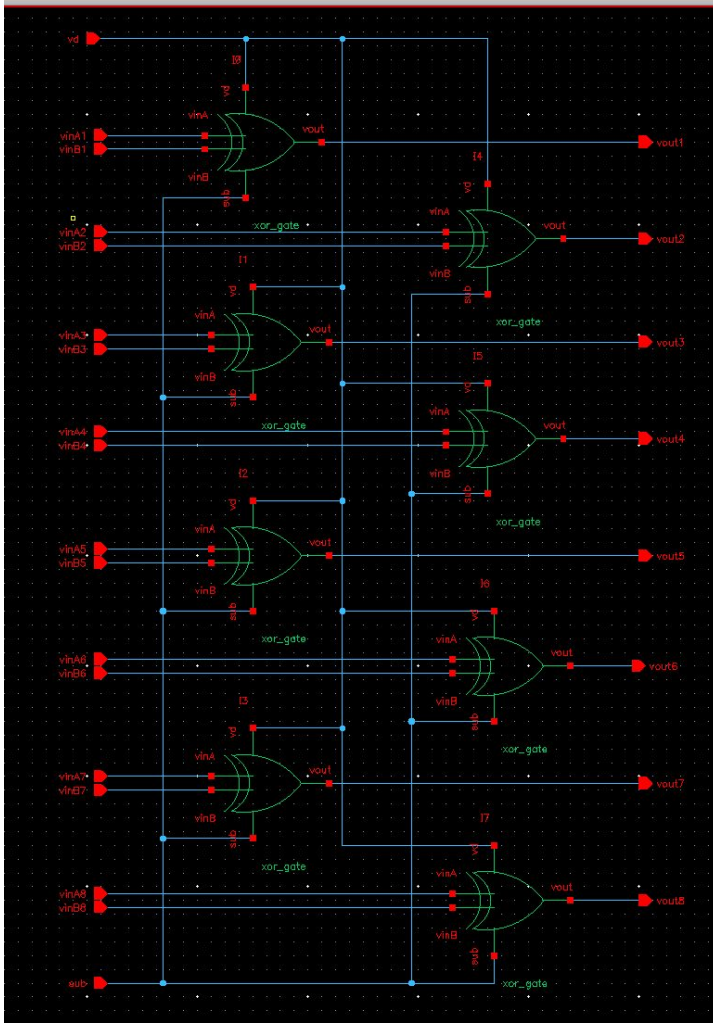


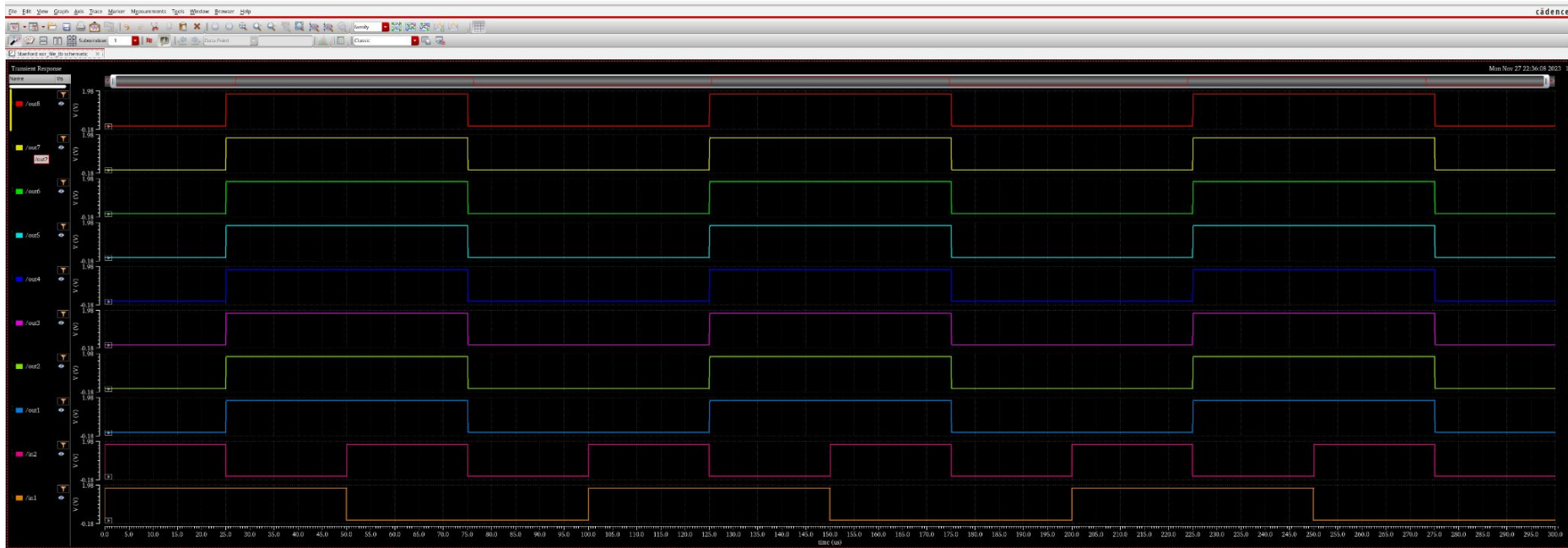
Xor - Gate Test Bench



Xor gate - Sim Shows the 2 input Xor Gate truth table

Xor File - Schematic



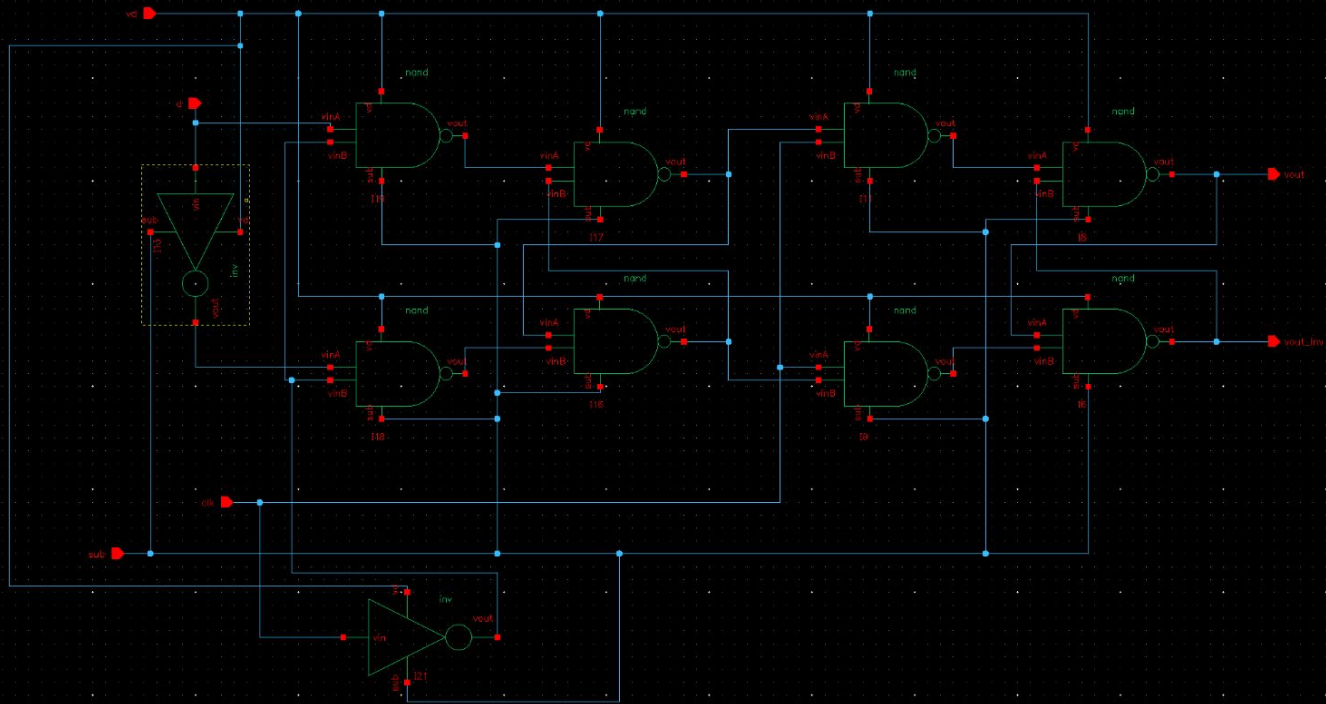


Xor File - Sim

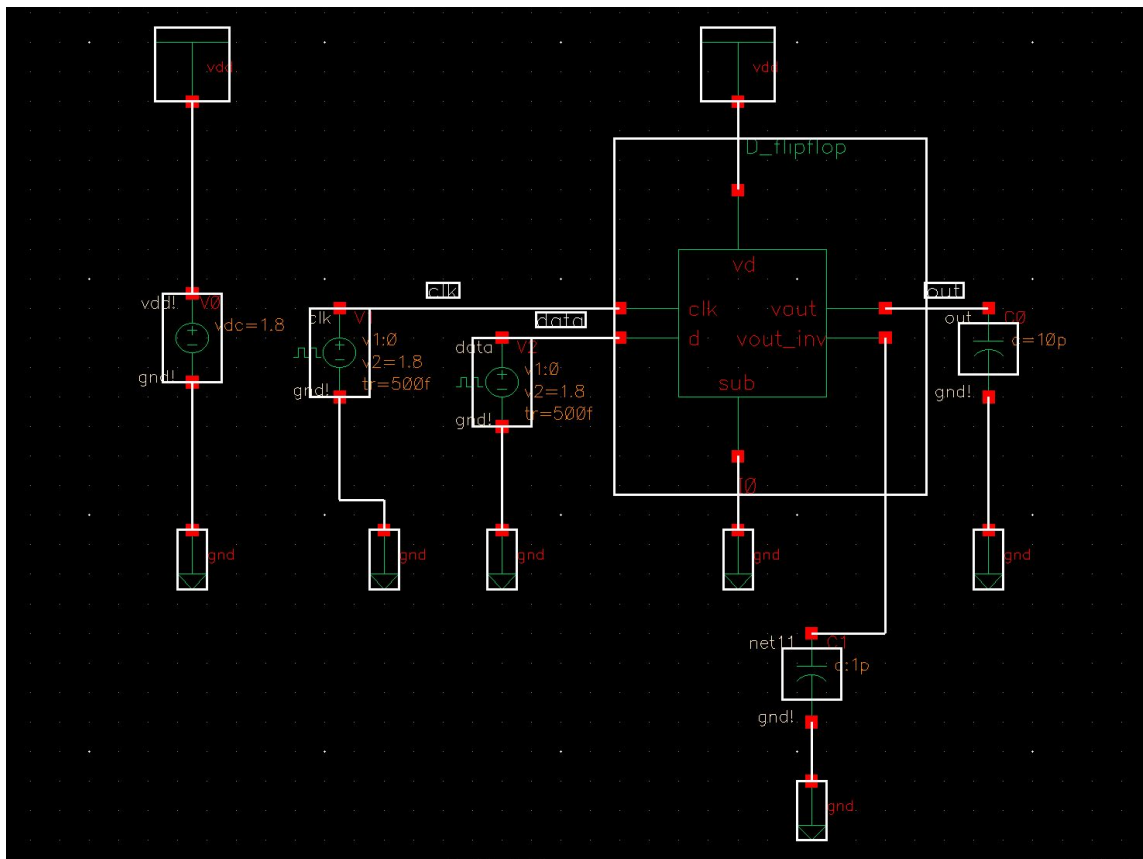
The Xor file is just 8 Xor gates in parallel. The Sim is identical to the xor gate sim



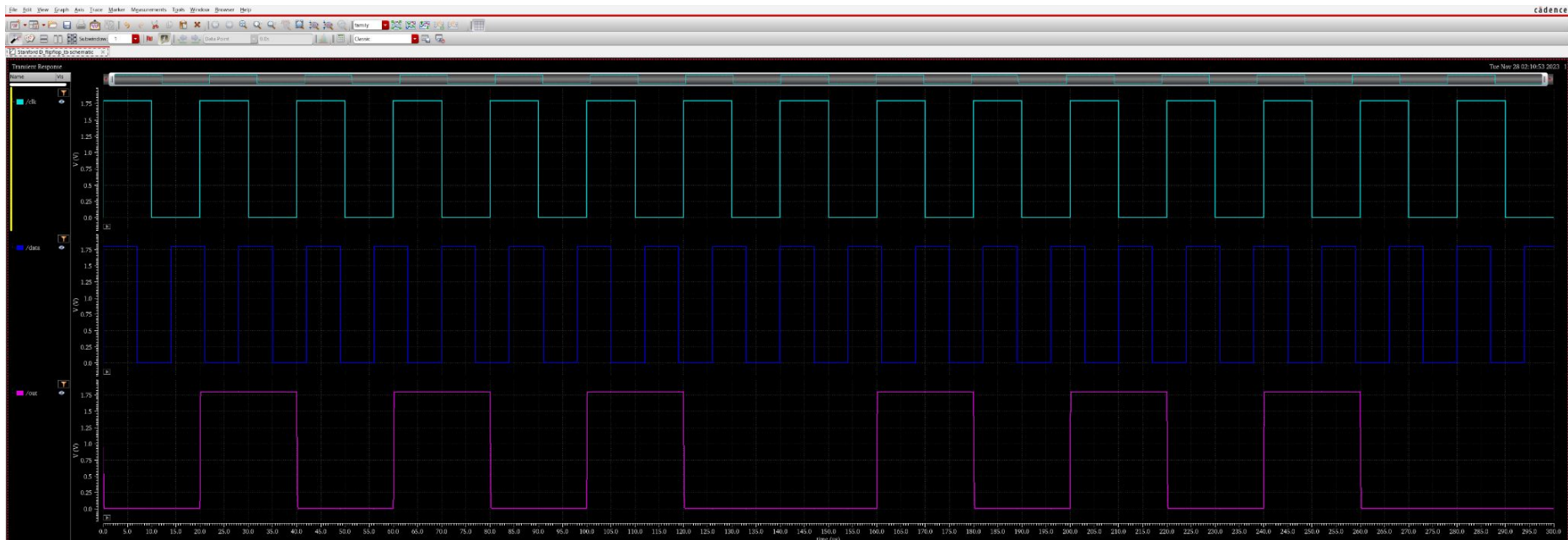
Storage



D- Flip Flop schematic

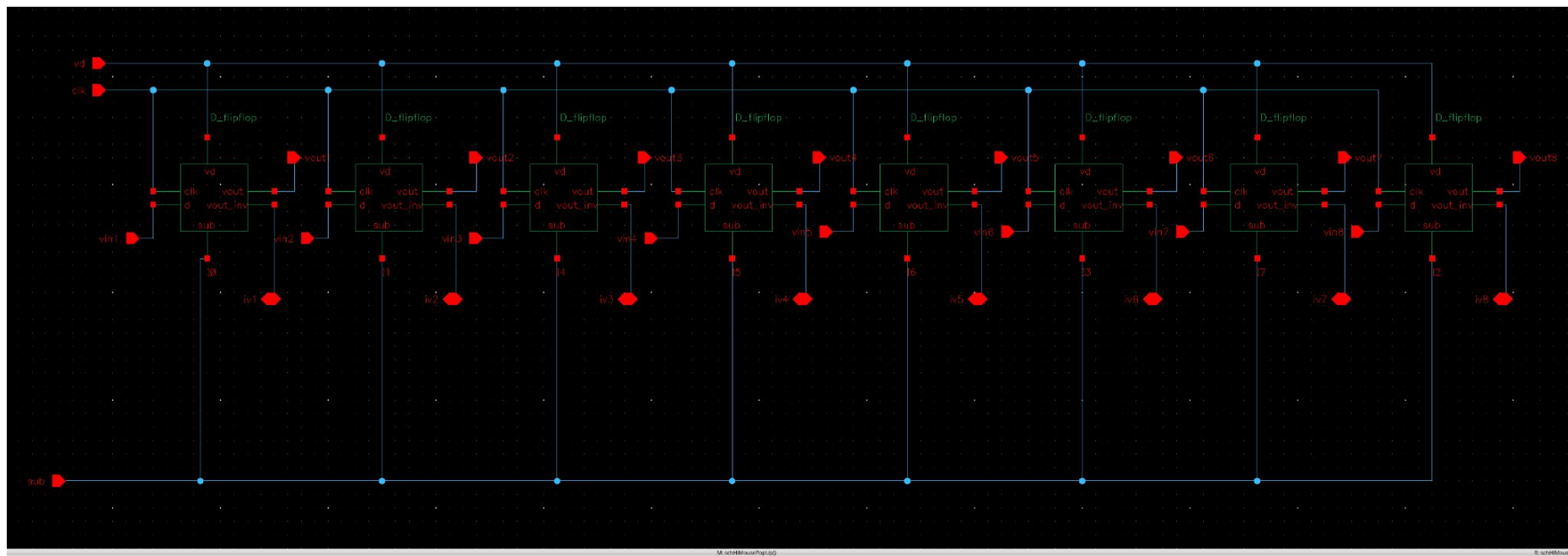


D- Flip Flop Testbench schematic

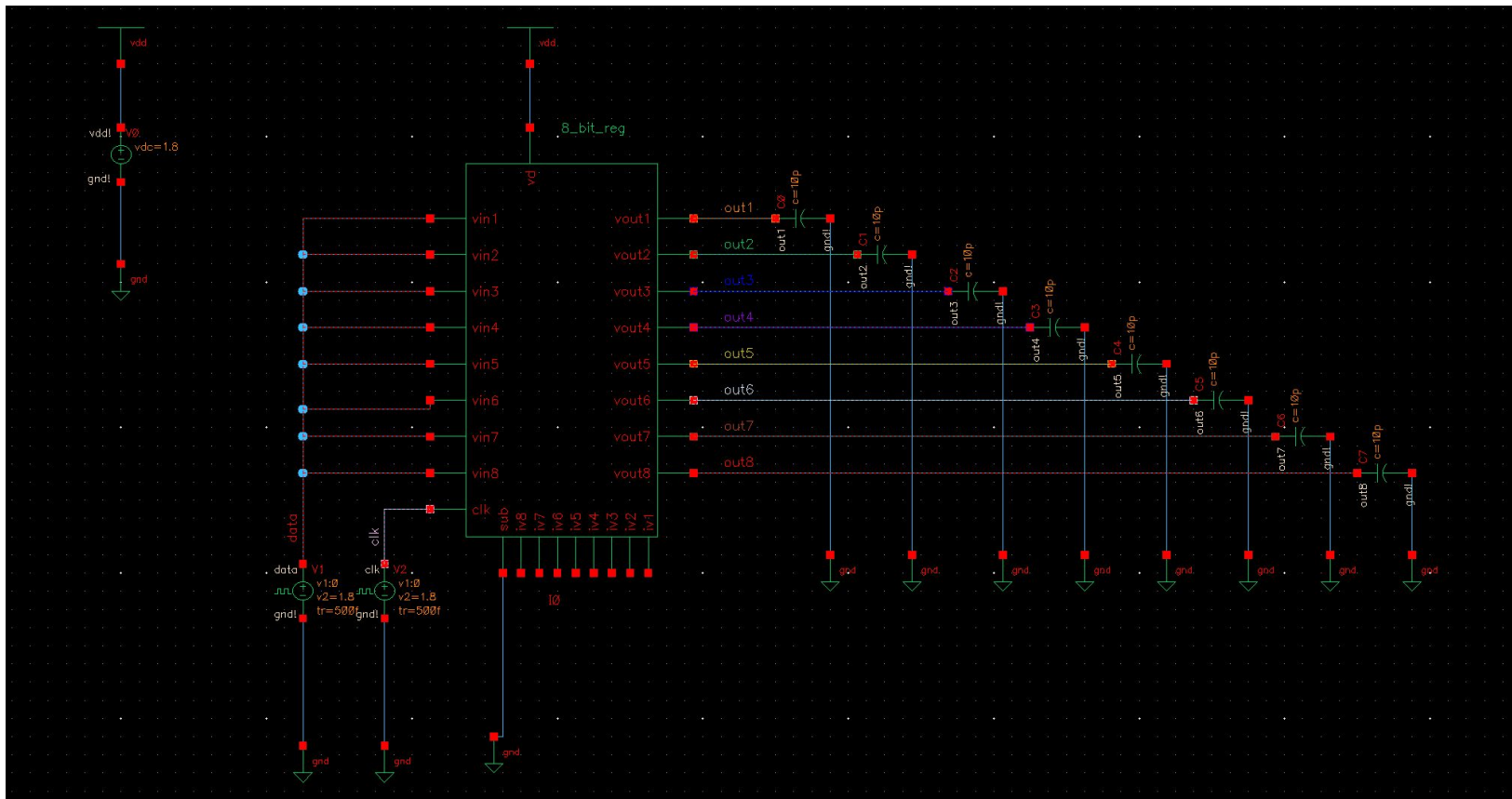


D- Flip Flop Testbench sim.

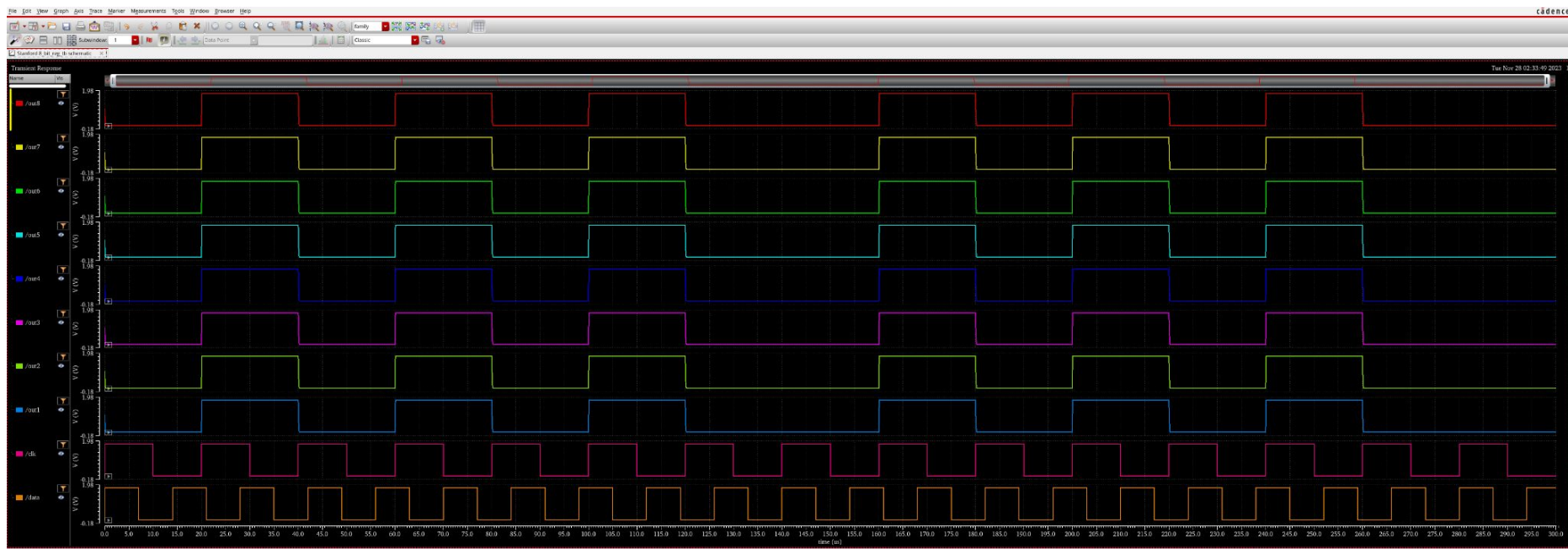
Since the Flip Flop is edge triggered, the data output at the edge is what is output. Sections where the Data and Clk rise at the same time output 0



8 bit register - Schematic



8 bit register - TestBench



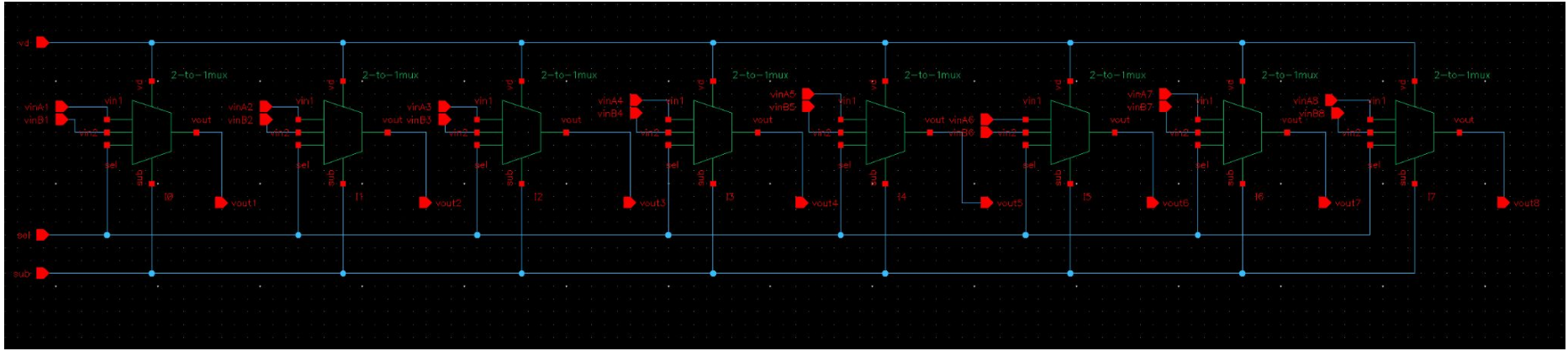
8 bit register - Sim

Like the Xor file, the 8 bit register is just 8 parallel flip flops, so the sim is identical to the single version.

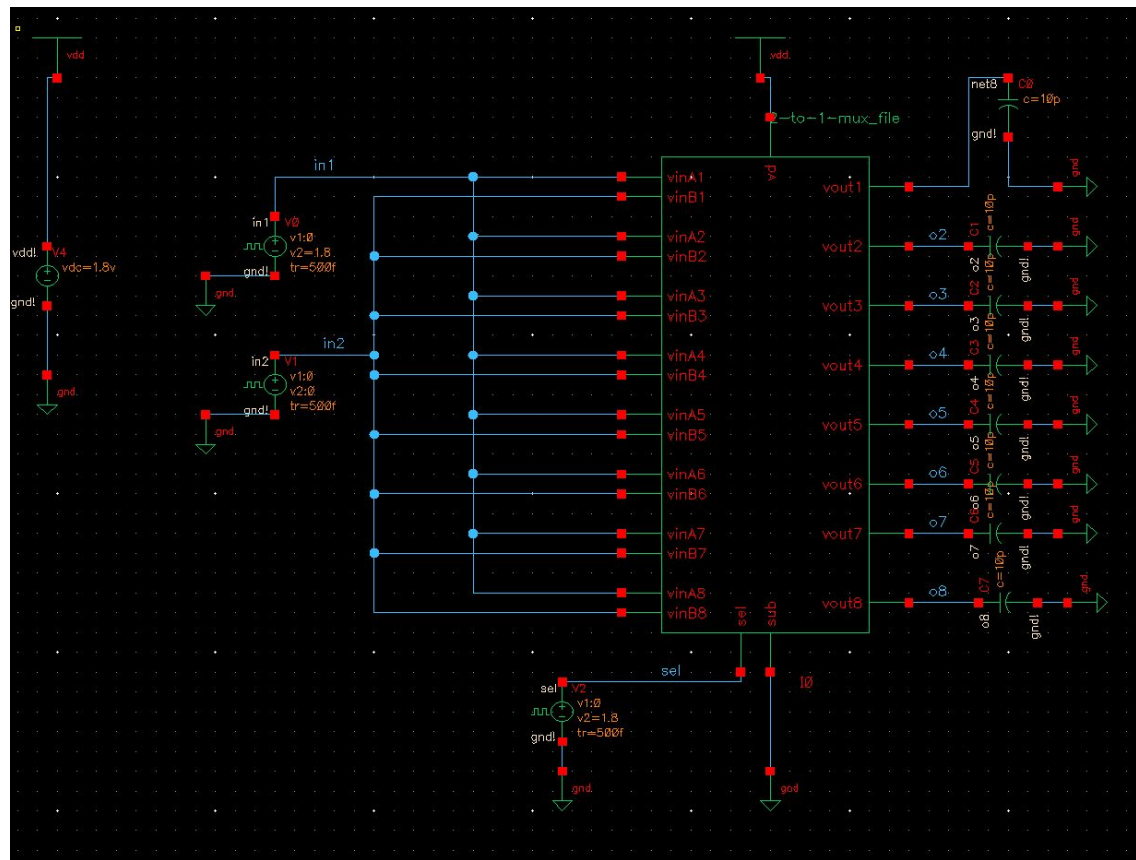


Decryption

Decryption uses another xor file and a 2-to-1 mux file for optional decryption.



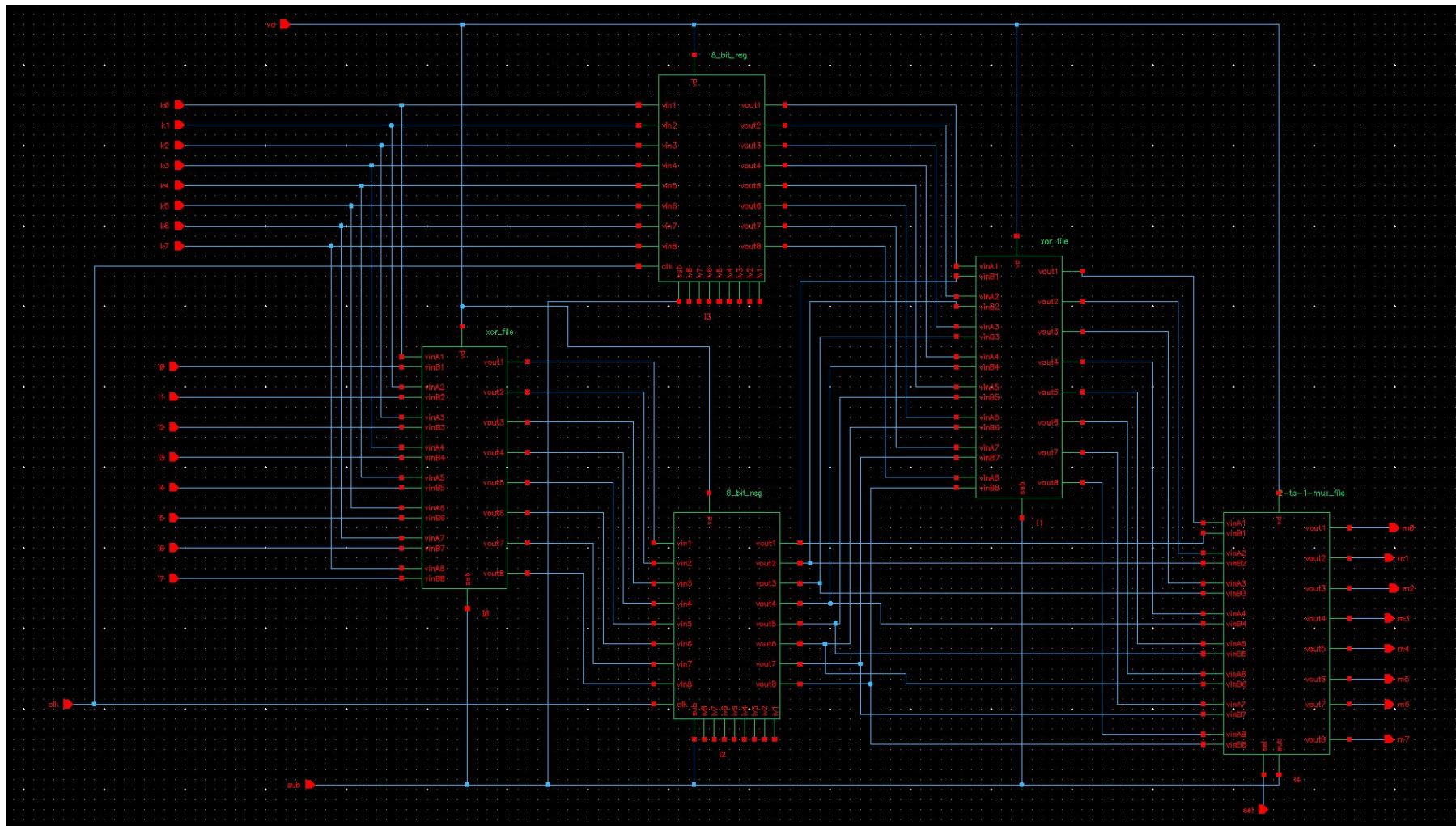
Mux-file - Schematic



Mux-file - Testbench Schematic



Encryptor / Decryptor





The Key inputs are at the top and are sent to the first xor file for encryption and the top most 8 bit register for storage.

The data inputs are lower and head into the xor file.

The encrypted data and key are stored in their own registers.

For decryption, the encrypted data and the same key are passed to the second xor file. Using the mux file, either the encrypted data or original version can be output.

